5. In 1985, a new allotrope of carbon was discovered and named buckminsterfullerene. This allotrope consists of sixty carbon atoms joined together to resemble a shape similar to that of a football (**Figure 1**), only ten septillion (10,000,000,000,000,000,000,000,000) times smaller.

This form of carbon was named after the architect Buckminster Fuller, famous for designing geodesic domes, as shown in **Figure 2**.

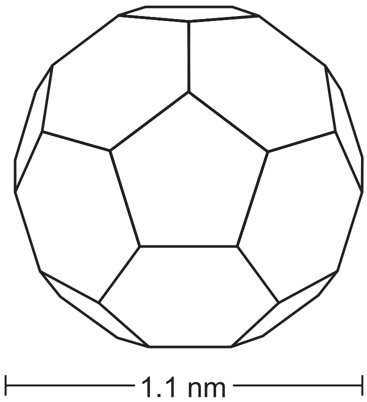

Figure 1

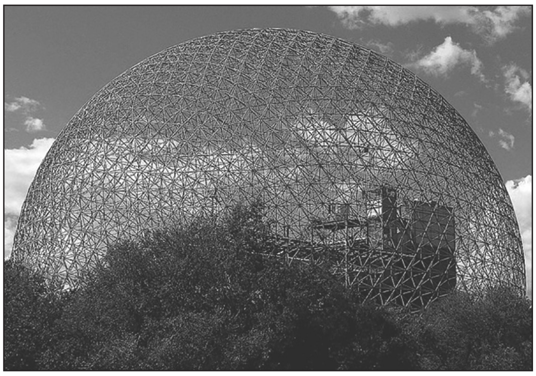

Figure 2

The structure of buckminsterfullerene is a truncated icosahedron, consisting of 20 hexagons and 12 pentagons that intersperse to form a spherical structure. Within the structure, each carbon atom is bonded to three other carbon atoms and no pentagon has a joining edge with another pentagon.

Other spherical allotropes of carbon, called fullerenes, have since been made. These include balls consisting of seventy, seventy-six and eighty-four carbon atoms. Together, they have become known as ‘Buckyballs’.

Fullerenes have high melting points and boiling points. They also have a high density and a large surface area for their size.

Today, fullerenes are at the heart of nanotechnology – the study of atomic scale structures and devices. This provides many exciting new research possibilities for scientists including their potential uses in catalysts, lubricants and in nano-tubes for strengthening materials and as a way of delivering drugs into the body.

Nano-tubes are fullerenes that are used to reinforce graphite in tennis rackets because they are very strong. They are also used as semiconductors in electrical circuits.

The nano-tube’s structure also allows it to be used as a container for transporting a drug in the body. A molecule of the drug can be placed inside the nano-tube cage. This keeps the drug ‘wrapped up’ until it reaches the site where it is needed. In this way, a dose that might be damaging to other parts of the body can be delivered safely to, for example, a tumour.

Examiner only

(a) Tick (✓) **all** the statements that correctly describe buckminsterfullerene. [2]

its structure has 32 faces	☐
it has a relative molecular mass of 60	☐
it is an allotrope of carbon	☐
it has a giant ionic structure	☐
it is a hydrocarbon compound	☐
it is a smart material	☐
it is 1×10^{25} times smaller than a football	☐

(b) (i) Use the following formula to calculate the number of edges that a molecule of buckminsterfullerene has. [2]

$$\text{number of edges} = \frac{\text{total number of sides of all pentagons and hexagons}}{2}$$

Number of edges = ...

Examiner only

(ii) Calculate, in m^3, the approximate volume of a buckminsterfullerene molecule. Write your answer to **three** significant figures and in standard form. [3]

$$\text{volume} = \frac{4}{3}\pi r^3$$

$\pi = 3.14$ r = radius

$1\,\text{nm} = 1 \times 10^{-9}\,\text{m}$

Volume = ... m^3

(c) Give the **main** reason why the structure of fullerenes has resulted in there being an interest in developing their use as catalysts. [1]

....

(d) State why some people might oppose the use of fullerenes in drug delivery systems in the body. [1]

....

....

(e) One student said that Buckyballs should be good electrical conductors but her friend disagreed.

Use your knowledge of bonding and structure to give **one** reason that **each** student could use to support their argument. [2]

....

....

....

....

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